ABSTRACT

Vehicle Plate Recognition (VPR) is the extraction of vehicle license plate information from an image. The system model uses already captured images for this recognition process. First the recognition system starts with character identification based on number plate extraction, Splitting characters and template matching. VPR as a real life application has to quickly and successfully process license plates under different environmental conditions, such as day time. It plays an important role in numerous real-life applications, such as automatic toll collection, traffic law enforcement, parking lot access control, and road traffic monitoring. The system uses different templates for identifying the characters from input image. After character recognition, an identified group of characters will be compared with database number plates for authentication. The proposed model has low complexity and less time consuming in terms of number plate segmentation and character recognition. This can improve the system performance and make the system more efficient by taking relevant samples. at the same time compared their advantages and disadvantages, which provide the basis for license plate recognition.

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LIST OF ABBREVIATIONS

ANN	-	Artificial Neural Networks
APSRTC	-	Andhra Pradesh State Road Transport Corporation
ASCII	-	American Standard Code for Information Interchange
AVI	-	Automatic Vehicle Identification
CPR	-	Car Plate Recognition
DLL	-	Dynamic Link Library
DTCNN	-	Discrete-Time Cellular Neural Network
DWT	-	Discrete Wavelet Transform
GMW	-	Globally Matched Wavelet
FIR	-	Filter's Impulse Response
HSR	-	High Security Registration
ITS	-	Intelligent Transport System
LNPR	-	License Number Plate Recognition
MEC	-	Maximum Entropy Criterion
OCR	-	Optical Character Recognition
PEE	-	Portion Edges Elimination
PIF	-	Proximity To The Image Frame
PNN	-	Probabilistic Neural Network
RGA	-	Region-growing algorithm
RTO	-	Regional Transport Office
SUSAN	-	Smallest Univalue Segment Assimilating Nucleus
VPR	-	Vehicle Range Plate Recognition

CHAPTER 1

INTRODUCTION

Vehicle number-plate recognition (VNPR) is a technology that uses optical character recognition on images to read vehicle registration plates to create vehicle detailed data. It can use existing closed-circuit television, road-rule enforcement cameras or cameras specifically designed for the task. VNPR is used by police forces around the world for law enforcement purposes, including checking if a vehicle is registered or licensed. Automatic number-plate recognition can be used to store the images captured by the cameras as well as the text from the license plate, with some configurable to store a photograph of the driver. System commonly use infrared lighting to allow the camera to take the picture at any time of day or night. VNPR technology must take into account plate variations from place to place. The software aspect of the system runs on standard home computer hardware and can be linked to other applications or databases. It first uses a series of image manipulation techniques to detect, normalize and enhance the image of the number plate and then character recognition to extract the alphanumeric of the license plate. VNPR systems are generally deployed in one of two basic approaches one allows for the entire process to be performed at the lane location in real-time and the other transmits all the images from many lanes to a remote computer location and performs the OCR process there at some later point in time.

CHAPTER 2

LITERATURE REVIEW

[1] Kodabagi, Vijayamahantesh, Kanavi (2011), 'An Approach for Automatic Vehicle License Plate Recognition Using Classification Algorithm', International Journal of Advanced Research in Engineering Technology

License Plate Recognition (LPR) has been intensively studied in many countries. Due to the different types of number plates being used, the requirements of an automatic number plate recognition system are different for each country. In general, objective of any LPR system is to localize and recognize potential license plate regions of the vehicle images captured from handheld device, digital camera, mobile phone camera. The remote computers then perform further operations like OCR on the stored images sent by the cameras at the lane-level. In order to process the high amount of images stored, a server farm is used which comprises of many computers working together. Localization of the license plate is done using morphological operations, horizontal and vertical edge processing. Character segmentation is carried out using connected component labeling. Character recognition is done by using Neural Network classifier. The method is tested on 100 samples of Indian vehicle images. The system achieves 86% accuracy in localization, 81% accuracy in segmentation and 80% high accuracy in character recognition.

[2] Pawan Wawage, Shraddha Oza (2013), 'An Approach for Automatic Detection of Vehicle License Plate and Character Recognition Using Classification Algorithm',International Journal of Emerging and technology and Advanced Engineering

Automatic Identification of Vehicle License Plate is a real time embedded system which identifies the characters directly from the image of the vehicle license plate. Since number plate guidelines are not strictly practiced everywhere, it often becomes difficult to correctly identify the non-standard number plate characters. The proposed method for License Plate Recognition (LPR) system works in three modules: localization of license plate, segmentation of the characters and recognition of the characters from the license plate. The main contribution of this paper is to provide a brief source of reference study of various License Plate Localization methods depending on the techniques they have used and a comparison of each technique in terms of their merits, demerits, computational power, detection success rate and accuracy, the execution time and speed of the whole License Plate System, which will be helpful for the researchers. This method is extremely useful for traffic police to search out the small print of an automobile violating the traffic rules. Its applications additionally include Automatic toll assortment systems and automobile parking systems. This paper deals with computing techniques from the field of Artificial Intelligence, machine vision and neural networks in construction of an Automatic System for Identification of Vehicle License Plate and Character Recognition.

[3] Ami Kumar Parida, SH Mayuri, Pallabi Nayk, Nidhi Bharti (2018), 'Recognition of Vehicle Number Plate Using MATLAB', Journal of University of Shanghai for Science and Technology

Automatic video analysis from traffic surveillance cameras is a fast-emerging field based on computer vision techniques. It is a key technology to public safety, intelligent transport system and for efficient management of traffic. Currently most reliable approach is through the recognition of number plates or radio frequency transponders. The automated variety Plate Recognition System plays a crucial role in addressing these problems as its application ranges from parking admission to observation urban traffic and to chasing automobile thefts. Their area unit varied ANPR systems accessible these days that area unit supported different methodologies. During this paper, we tend to commit to reviewing the assorted techniques and their usage. The automated variety Plate Recognition System plays a crucial role in addressing these problems as its application ranges from parking admission to observation urban traffic and to chasing automobile thefts. Proposed two methods for extraction of license plates and comparing it with other existing methods. The Extracted license plates are segmented into individual characters by using a region-based method. The recognition scheme combines adaptive iterative thresholding with a template matching algorithm

[4] Narendra Singh Tomar, Prakhar Sachan, Pranav Mittal, Shivani Agarwal (2020), 'Vehicle range plate detection victimization Matlab', International Journal of Science Research and Engineering Trend.

The Vehicle range Plate Recognition (VPR) system is predicated using image process technology. It is one of the necessary systems designed to notice the vehicle range plate. In today world with the increasing range of cars day by day, it is impractical to manually keep a record of the complete vehicle. With we this method, it becomes straightforward to stay a record and use it whenever needed. The most objective here is to style associate degree economical automatic vehicle identification system by victimization vehicle range plate. The system first would capture the image of the vehicle as shortly because the vehicle reaches the safety checking space. Its applications additionally include Automatic toll assortment systems and automobile parking systems. In high-security areas wherever the parking lot is reserved for VIP vehicle house owners solely, the parking gate is opened when range recognition. The captured pictures area unit is then extracted by the victimization of the segmentation method. Optical character recognition is employed to spot the characters. The captured pictures area unit is then extracted by the victimization of the segmentation method. Optical character recognition is employed to spot the characters. The obtained information is then compared with the information kept in their info. The system is enforced and simulated on MATLAB and performance is tested on real pictures. This kind of system is widely utilized in control areas, tolling, park etc. This method is principally designed for the aim of the security system.

CHAPTER 3

EXISTING SYSTEM

The system for automatic car license plate recognition includes a computer, and custom designed software for image processing, analysis and recognition. Optical Character Recognition (OCR) technique was used, which is a widely used technology which translates scanned images of printed text into machine encoded text. Artificial Neural Networks (ANN) are widely used intelligent computing architecture for pattern recognition. The size corresponding to the best suitable result is used for the final neural network. In order to process the high amount of images stored, a server farm is used which comprises of many computers working together. The developed system first detects the vehicle and then captures the vehicle image. Vehicle number plate region is extracted using the image segmentation in an image. Common pattern matching technique is a simple technique for the recognition of single font and fixed size character, which is a suitable approach for ANPR systems. The remote computers can be linked with the database which stores the details of the car owners and thus the required information can be obtained. The best possible number plate location is found out by comparing width by height factor of actual Indian number plates to the same factor of plate like areas found by this method.

CHAPTER 4

PROPOSED SYSTEM

The proposed system is intended to address two key issues that are common in educational institutions, including keeping track of the quantity and type of vehicles with the precise time their vehicle departed the premises in the event of thefts. The proposed system block diagram is as shown in fig.5.1.

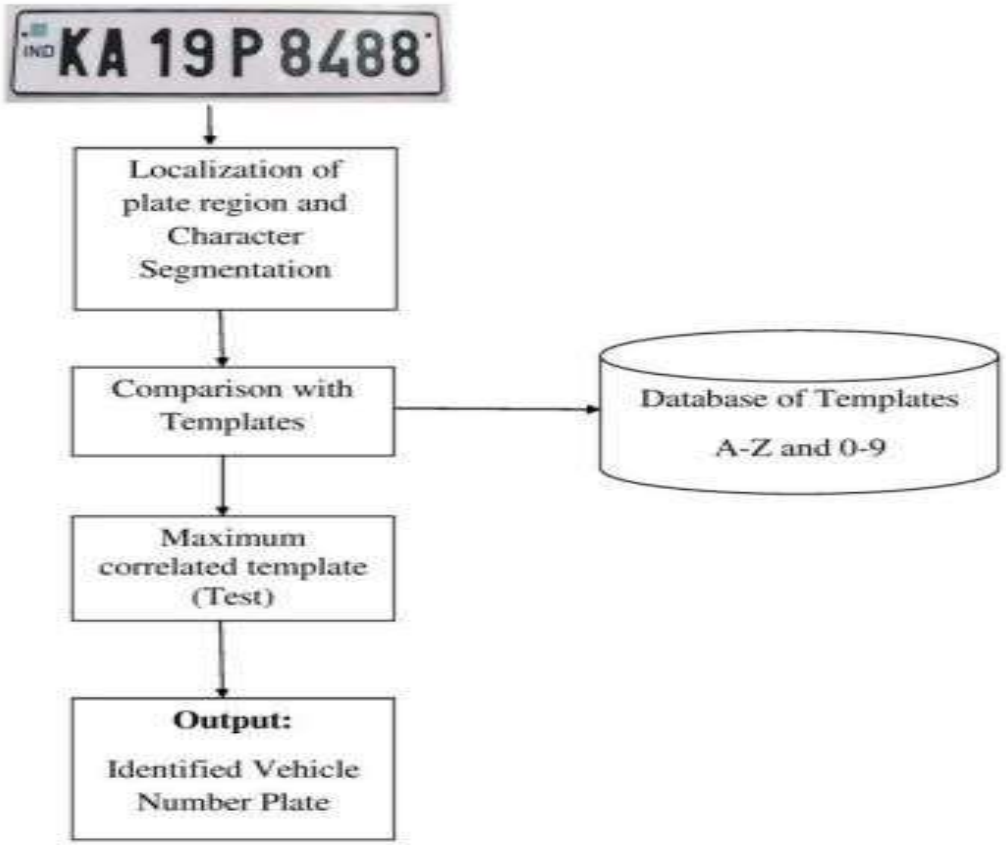


Figure 5.1 Block diagram of system for vehicle number plate recognition using Template Matching.

CHAPTER 5

HARDWARE AND SOFTWARE REQUIREMENTS

This proposed work of "Vehicle plate recognition using MATLAB" will use both hardware and software. To detect the number plate and to analyze the character segmentation and it uses the MATLAB software.

5.1 HARDWARE REQUIREMENTS

- PC Computer (or) Laptop

5.2 SOFTWARE REQUIREMENTS

- ✓ Operating system : Windows XP (or) Windows 10
- ✓ SOFTWARE : MATLAB R2015a

5.2.1 SOFTWARE DESCRIPTION

✓ SOFTWARE : MATLAB R2015a

TOOLBOX	VERSION
Matlab	8.5
Simulink	8.5
Communication_toolbox	6.0
Control_fitting_toolbox	9.9
Curve_fitting_toolbox	3.5.1

Dsp system toolbox	9.0
Financial toolbox	5.5
Fixed_point_designer	5.0
Fuzzy_Logic_Toolbox	2.2.21
Image_Processing_Toolbox	9.2
Compiler	6.0

MATLAB R2015a (version 8.5) is installed in SERC. MATLAB, the language of technical computing is a programming environment for algorithm development, data analysis, visualization and numeric computation. Simulink is a graphical environment for simulation and Model-Based Design of multidomain dynamic and embedded systems. MATLAB parallel profile works only on the local system.

MATLAB is an interactive programme with on-line utilization. To get an overview of MATLAB type intro from the MATLAB prompt. The help command tells you the topics (directories) available in MATLAB. The command help directory name or help function name gives more detailed help about the directory or the function.

CHAPTER 6

SYSTEM DESIGN

6.1 SYSTEM BLOCK DIAGRAM

In this system, the plate detected from Captured image and convert the colour image into gray scale image then the convert image is segmented by character. Then the Open Character Recognition (OCR) converts the image of text into a machine readable text format. The American Standard Code for Information Interchange (ASCII)is represents both control characters and printable characters. Figure 6.1 shows the overall block diagram of the system.

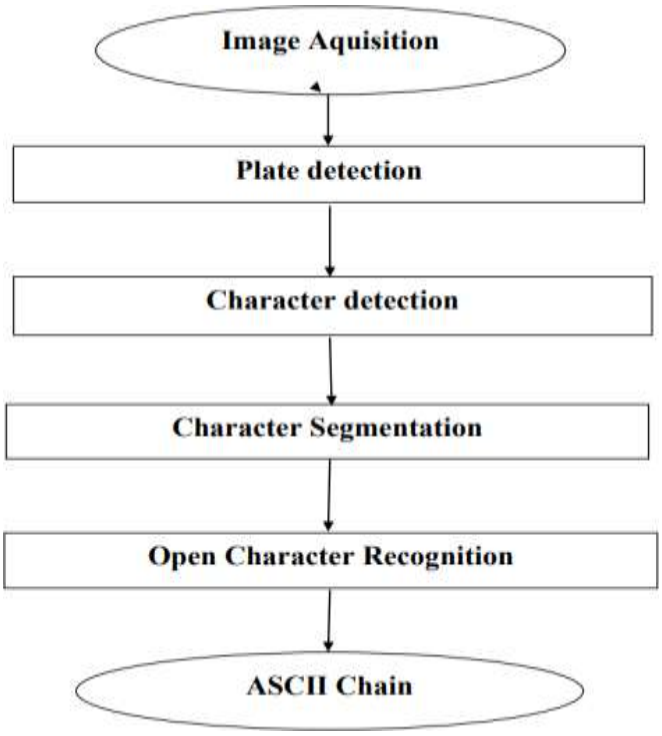


Figure 6.1 Block Diagram of Proposed System

6.1.1 IMAGE ACQUISITION

Image acquisition is the first step in image processing. This step is also known as preprocessing in image processing. It involves retrieving the image from a source, usually a hardware-based source.

6.1.2 PLATE DETECTION

This approach uses the image of the car to localize the prospective licence plate. The is algorithm is in charge of location and separating the licence plate from the rest of the picture of the car. The car plate's measurements are corrected for skew and adjusted to the necessary size.The plate is scaled down to standard size. The visual representation of the cars is free of noise.The algorithm is used to change the images contrast and brightness.

6.1.3 CHARACTER DETECTION

The presence of characters can be used to find the correct location of the number plate on the image of the vehicle. In this technique, first of all, the presence of text on the image is searched. The region having the presence of characters is extracted as the potential licence plate from the image of the vehicle. These methods are robust to the rotation but time-consuming.

6.1.4 CHARACTER SEGMENTATION

The goal of this stage, given the dilation image, is to segment each character while maintaining their unique characteristics. Segmentation is one of the most important steps in automatic number plate recognition. If segmentation fails, a character could be incorrectly divided into two parts or mixed up with another character. To later distinguish the characters on the car number plate, each character must be suitably spaced apart. The various

characters must be distinguished from one another (segmented). The characters and numbers on the licence plate are divided using character segmentation, and each section is saved as a different image.

6.1.5 OPEN CHARACTER RECOGNITION

One method for character recognition is template matching. The template technique is used to locate a sub image within an image. This process makes use of a database of characters, or templates. The Regular font type is used to produce templates for each of the alphanumeric characters. The current input character is compared to each template to identify either an exact match or the template that most closely resembles the input character before recognition may take place.

By repositioning the standard template, it can capture the ideal location in which the character is and perform an accurate match. Based on the target characters template, the moving template matching method compares the target character to the standard character's template in eight different directions: up, down, left, right, upper left, lower left, upper right, and lower right. Open character recognition (OCR), which compares each character to the entire alphanumeric database, is another technique for character recognition. In order to match each individual character, OCR really uses the correlation method. Once the number has been located, it is then saved in a variable in string format. According to the comparison's outcome, the resultant signals are presented. There will be templates for all the characters, from A to Z and 0 to 9.

6.1.6 AMERICAN STANDARD CODE FOR INFORMATION INTERCHANGE (ASCII)

One popular technique for representing text on computers is the ASCII (American Standard Code for Information Interchange) encoding scheme. Seven bits, or one unique code or integer ranging from 0 to 127, are used to represent each ASCII character. These codes are made up of letters, numbers, punctuation marks, and special characters, including line breaks, tabs, and control characters. Chain coding is based on identifying and storing the directions from each pixel to its neighbor pixel on each contour. Before defining this process, it is necessary to clarify the various types of neighbors that are associated with a given pixel in a binary image. Because ASCII only uses 8 bits, one significant drawback is that it can only represent 256 different characters. As a result, the various character types used in languages around the world cannot be encoded using ASCII.

The committee eventually decided on a 7-bit code for ASCII. 7 bits allow for 128 characters. While only American English characters and symbols were chosen for this encoding set, 7 bits meant minimized costs associated with transmitting this data (as opposed to say, 8 bits). The first 32 characters of ASCII were reserved control characters. These characters were used to relay special instructions to other devices, like printers. ASCII offers a universally accepted and understood character set for basic data communications. It enables developers to design interfaces that both humans and computers understand. ASCII codes a string of data as ASCII characters that can be interpreted and displayed as readable plain text for people and as data for computers.

CHAPTER 7

WORKING PROCESS

7.1 THE GENERAL NPR SYSTEM

7.1.1 Vehicle Image Captured

The image of the vehicle whose number plate is to be identified is captured using digital camera of 3.2 megapixel.



Figure 7.1 Captured Image vehicle

7.1.2 Extraction of Number Plate

In this process the number plate is extracted by converting RGB Image i.e., the captured image to Gray Scale Image. Here mathematical morphology is used to detect the region and Sobel operator are used to calculate the threshold value. Then imfill function is used to fill the holes so that we get a clear binary image.

7.1.3 Segmentation and Recognition of Plate Character

Here bounding box technique is used for segmentation. The bounding box is used to measure the properties of the image region. The basic step in recognition of vehicle number plate is to detect the plate size. Here the segmented image is multiplied with gray scale image so that only get the number plate of the vehicle.

7.1.4 Display Vehicle Number

After undergoing the above steps the number plate is displayed in MATLAB window.

7.2 NPR IMPLEMENTATION USING MATLAB

7.2.1 INPUT IMAGE

This is the first phase deals with acquiring an image. In proposed system, digital camera of 3.2 megapixel camera is used. The input image is 120 x160 or 1200 x 1600.

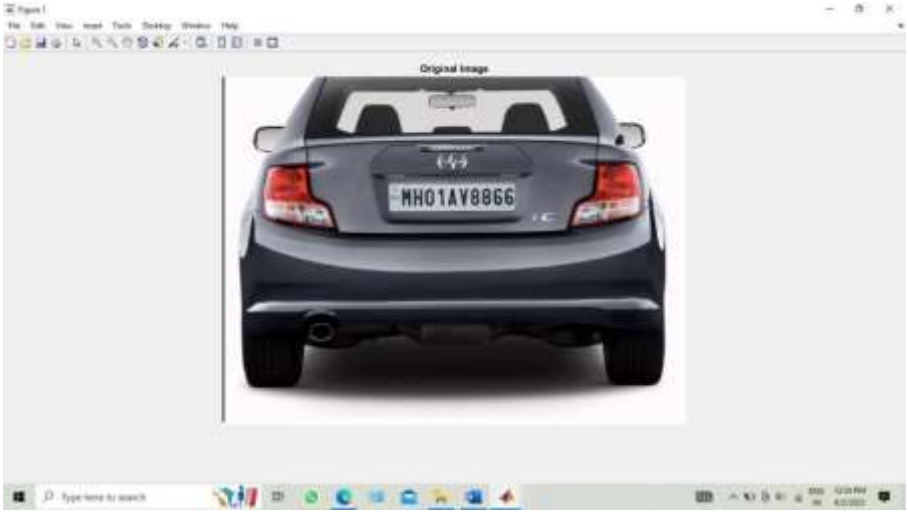


Figure 7.2 Original Image

7.2.2 EXTRACTION OF NUMBER PLATE LOCATION

The inputs to the system were the images of vehicles captured by a camera. RGB to gray-scale conversion is adopted, in order to facilitate the plate extraction and increase the processing speed. Colour image (RGB) acquired by a digital camera is converted to gray-scale image using

I gray=0.114*R+0.587*G+0.299*B……

The basic step in recognition of vehicle number plate is to detect the plate size. In general number plates are in rectangular shape, hence it is necessary to detect the edges of the rectangular plate. The binary gradient mask shows lines of high contrast in the image. These lines do not quite delineate the outline of the object of interest. Compared to the original image, gaps in the lines are observed that surrounds the object in the gradient mask. These linear gap disappears if the image is dilated using linear structuring elements. Structuring element is represented as matrices, which is a characteristic of certain structure

and features to measure the shape of an image which is used to carry out other image processing operations. The binary gradient mask is dilated using the vertical structuring element followed by the horizontal structuring element.

After removal of lower than 100 connected pixels. The dilated gradient mask shows the outline of the region quite nicely, but there are still holes in the interior of the region, to fill these holes imfill function in MATLAB is used. By filling holes the image of the captured vehicle.

7.2.3 REMOVE CONNECTED OBJECTS ON BORDER

The region of interest has been successfully segmented, but it is not the only object that has been found. Any objects that are connected to the border of the image can be removed using the imclear border MATLAB function. The connectivity in the function was set to 4 or 8 to remove diagonal connections and fill the hole to locate the plate region. After removing the lower pixel components actual plate region is detected.

Finally, in order to make the segmented object look natural, the image is eroded twice with one of the diamond, disk and line structuring element. This helps in extraction of number plate area of the vehicle.

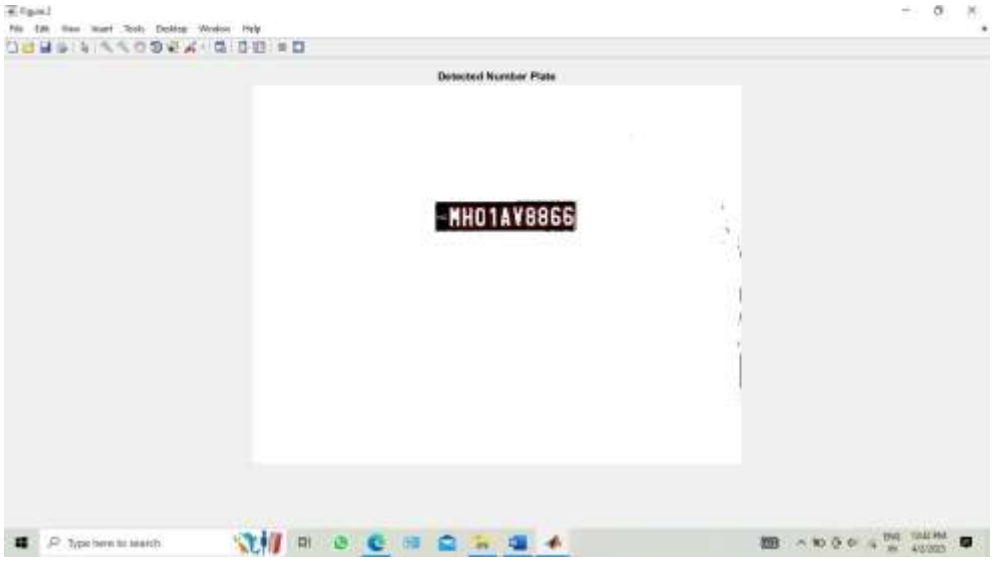


Figure 7.3 Number Plate with bounding box image

7.2.4 CHARACTER RECOGNITION & DISPLAY THE RESULT

It is employed for the purpose of conversion of images of text into characters. Number plate recognition is now used to compare the each individual character against the complete alphanumeric database using template matching. The Regular font type is used to produce templates for each of the alphanumeric characters. The task of binarisation itself is necessary since most commercial recognition algorithms work only on binary images since it proves to be simpler.

Recognition is one of the most important processes in the number plate recognition, because all further steps rely on it. If the segmentation fails, a character can be improperly divided into two pieces, or two characters. The ultimate solution on this problem is to use bounding box technique. Once a bounding box created over each character and numbers presented on number plate, each character & number is separate out for recognition of number plate.

The matching process moves the template image to all possible positions in a larger source image and computes a numerical index that indicates how well the template matches the image in that position. It compares the input image with every image in the template to get the match. Edge detection is a part of segmentation used for finding the boundary of an object or region within an image. It helps in data extraction. Matching is done on a pixel by pixel basis. Since the template size is fixed, it leads to accurate recognition. It displays number plate of the desired vehicle.

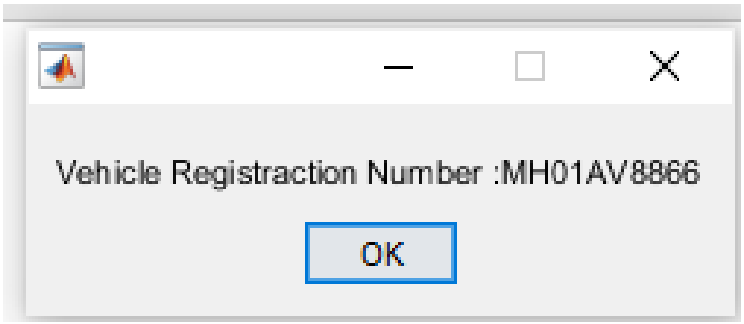


Figure 7.4 Image of Result

CHAPTER 8

TESTING

8.1 TESTING DESCRIPTION

- Find all the contours in the image.
- Find the bounding rectangle of every contour.
- Compare and validate the sides ratio and area of every bounding rectangle with an average license plate.
- Apply image segmentation in the image inside the validated contour to find characters in it.
- Recognize characters using an OCR.

Methodology:

1. To reduce the noise we need to blur the input Image with Gaussian Blur and then convert it to grayscale. Find vertical edges in the image. To reveal the plate we have to binarize the image.
2. For this apply Otsu's Thresholding on the vertical edge image. In other thresholding methods, we have to choose a threshold value to binarize the image but Otsu's Thresholding determines the value automatically.
3. Apply Closing Morphological Transformation on the thresholded image. Closing is useful to fill small black regions

between white regions in a thresholded image. It reveals the rectangular white box of license plates.

4. To detect the plate we need to find contours in the image. It is important to binarize and morph the image before finding contours so that it can find a more relevant and less number of contours in the image. If you draw all the extracted contours on the original image.
5. Now find the minimum area rectangle enclosed by each of the contours and validate their side ratios and area. We have defined the minimum and maximum area of the plate as 4500 and 30000 respectively.
6. Now find the contours in the validated region and validate the side ratios and area of the bounding rectangle of the largest contour in that region.
7. After validating you will get a perfect contour of a license plate. Now extract that contour from the original image.

The LPRST evaluation protocol is designed to assess the state of the art of LPR, advance the existing technology and suggest future research directions. To succeed at this, the evaluation design must solve the 'three bears' problem the test cannot be too hard or too easy. If the test is too easy, the testing process becomes an exercise in 'tuning' existing algorithms. If the test is too hard, the test is beyond the ability of existing techniques and the results from the test are poor and do not allow for an accurate assessment of algorithmic capabilities.

The solution to the three bears problem is through the selection of images used in the evaluation and the evaluation protocol. Tests are administered using

an evaluation protocol that states the mechanics of the tests and the manner in which the test will be scored. In LPR, the protocol states the testing objectives, what images are selected for each object of the test and how the performance results are scored.

The characteristics and quality of the images are major factors in determining the difficulty of the problem being evaluated. For example, if the image resolution of the license plate is low, the problem is different from that for images in which the resolution of the license plate is high. In the license plate image database, variability is introduced by the inclusion of different resolutions, illuminations, angles, and so on, as described in Section .

The test is designed to measure the performance of algorithms or products that can automatically locate and recognise license plates from a database. The test is designed to measure technology performance and it is not concerned with real-time implementation issues such as the speed of implementation. This means there is a requirement for the quality of the testing images, that is, the images must contain a definite testing label, which is closely related to the technical testing elements. This is why we must quantify the image attributes and give each image a unique test label to provide a clear evaluation of the technical elements being measured.

CHAPTER 9

RESULTS AND DISCUSSION

9.1 OUTPUTS



Figure 9.1 Original Image Detection.



Figure 9.2 Detected Number Plate.

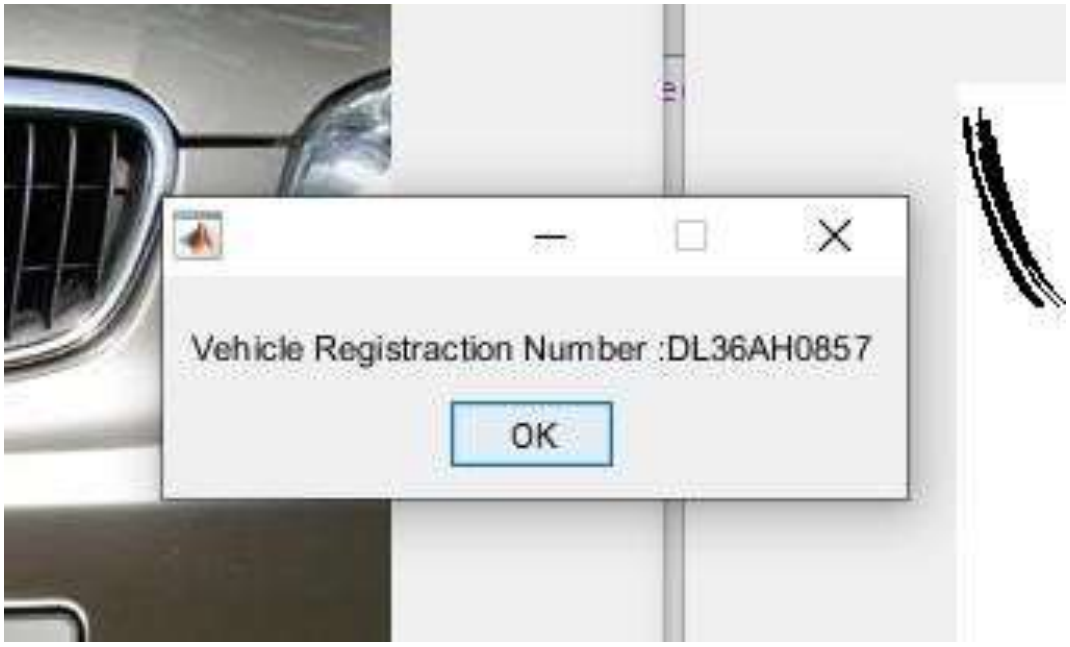


Figure 9.3 Vehicle Registration Number.

CHAPTER 10

CONCLUSION

The image segmentation problem in license plate recognition for Indian vehicle Plates has been examined in two stages: license plate Localization and extraction from the scene followed by the separation of the characters from the previously extracted license plate region background. Various approaches for license plate detection in an image, and extensive experiments have been devised to test them are presented. The purpose of this thesis has been to investigate the scope of automatic vehicle plate recognition under minimal restrictions. The main objective is aimed at contributing towards the research in the fields of machine vision, pattern analysis and image processing. The system developed investigates the possibility of automating the whole process of license plate recognition for a wide range of environments. Recognition has been an area of research for more than three decades now. Plate Recognition is praise-worthy. Variations in licence plates are one prominent problem and achieving a high degree of recognition accuracy is a tedious task. This entire research effort has been executed in a step-by-step fashion Given an input image, the system extract extracts the license plate, isolates the characters, and finally identify the characters. For each task, a set of methods are proposed, designed and developed. A better performing class of segmentation algorithms in comparison with the available algorithms in the field is developed. Experimental results showed that the proposed technique gives 96.69 % accurate results for character regions recognition.

APPENDICES

```
[path file] = uigetfile('*.jpg','Select Image');
img = imread(fullfile(file,path));
figure;
imshow(img);
title('Original Image');
img2 = rgb2gray(img);
imgX = edge(img2,'sobel','horizontal');
imgY = edge(img2,'sobel','vertical');
img3 = imgX|imgY;
img3 = bwareaopen(img3, 700);
se = strel('line',2,90);
img3 = imdilate(img3,se);
se = strel('line',2,0);
img3 = imdilate(img3,se);
img4 = imfill(img3,'holes');
img4 = bwmorph(img4,'thin',1);
se = strel('line',5,90);
img5 = imerode(img4,se);
img5 = imerode(img5,se);
img6 = immultiply(img2,img5);
verticalProfile = sum(img6, 2);
row1 = find(verticalProfile, 1, 'first');
row2 = find(verticalProfile, 1, 'last');
horizontalProfile = sum(img6, 1);
column1 = find(horizontalProfile, 1, 'first');
column2 = find(horizontalProfile, 1, 'last');
img7 = img6(row1-1:row2+1, column1-1:column2+1);
img7 = ~im2bw(img7);
figure;
imshow(img7);
title('Detected Number Plate');
imgProp = regionprops(img7,'BoundingBox','Image');
x=1;
im0=imread('C:\Users\HP\Desktop\Numbers_jpg\0.jpg');
im1=imread('C:\Users\HP\Desktop\Numbers_jpg\1.jpg');
im2=imread('C:\Users\HP\Desktop\Numbers_jpg\2.jpg');
im3=imread('C:\Users\HP\Desktop\Numbers_jpg\3.jpg');
im4=imread('C:\Users\HP\Desktop\Numbers_jpg\4.jpg');
im5=imread('C:\Users\HP\Desktop\Numbers_jpg\5.jpg');
```

```
im6=imread('C:\Users\HP\Desktop\Numbers_jpg\6.jpg');
im7=imread('C:\Users\HP\Desktop\Numbers_jpg\7.jpg');
im8=imread('C:\Users\HP\Desktop\Numbers_jpg\8.jpg');
im9=imread('C:\Users\HP\Desktop\Numbers_jpg\9.jpg');

imA=imread('C:\Users\HP\Desktop\Aphabets_jpg\A.jpg');
imB=imread('C:\Users\HP\Desktop\Aphabets_jpg\B.jpg');
imC=imread('C:\Users\HP\Desktop\Aphabets_jpg\C.jpg');
imD=imread('C:\Users\HP\Desktop\Aphabets_jpg\D.jpg');
imE=imread('C:\Users\HP\Desktop\Aphabets_jpg\E.jpg');
imF=imread('C:\Users\HP\Desktop\Aphabets_jpg\F.jpg');
imG=imread('C:\Users\HP\Desktop\Aphabets_jpg\G.jpg');
imH=imread('C:\Users\HP\Desktop\Aphabets_jpg\H.jpg');
imI=imread('C:\Users\HP\Desktop\Aphabets_jpg\I.jpg');
imJ=imread('C:\Users\HP\Desktop\Aphabets_jpg\J.jpg');
imK=imread('C:\Users\HP\Desktop\Aphabets_jpg\K.jpg');
imL=imread('C:\Users\HP\Desktop\Aphabets_jpg\L.jpg');
imM=imread('C:\Users\HP\Desktop\Aphabets_jpg\M.jpg');
imN=imread('C:\Users\HP\Desktop\Aphabets_jpg\N.jpg');
imO=imread('C:\Users\HP\Desktop\Aphabets_jpg\O.jpg');
imP=imread('C:\Users\HP\Desktop\Aphabets_jpg\P.jpg');
imQ=imread('C:\Users\HP\Desktop\Aphabets_jpg\Q.jpg');
imR=imread('C:\Users\HP\Desktop\Aphabets_jpg\R.jpg');
imS=imread('C:\Users\HP\Desktop\Aphabets_jpg\S.jpg');
imT=imread('C:\Users\HP\Desktop\Aphabets_jpg\T.jpg');
imU=imread('C:\Users\HP\Desktop\Aphabets_jpg\U.jpg');
imV=imread('C:\Users\HP\Desktop\Aphabets_jpg\V.jpg');
imW=imread('C:\Users\HP\Desktop\Aphabets_jpg\W.jpg');
imX=imread('C:\Users\HP\Desktop\Aphabets_jpg\X.jpg');
imY=imread('C:\Users\HP\Desktop\Aphabets_jpg\Y.jpg');
imZ=imread('C:\Users\HP\Desktop\Aphabets_jpg\Z.jpg');

cnt=0;
out='';
for i = 1 : size(imgProp)
    this = imgProp(i).BoundingBox;
    if(this(3)>=15 &&this(4)>=30 && this(3)<=80 &&this(4)<=160)
    temp = ~imgProp(i).Image;
    temp=imresize(temp,[40,20]);
    imwrite(temp,'C:\Users\HP\Desktop\temp_jpg\temp.jpg');
    imgTest =imread('C:\Users\HP\Desktop\temp_jpg\temp.jpg');
```

```
img = imabsdiff(imA,imgTest);
rectangle('position',this,'EdgeColor','r');
minDiff = sum(img(:));
cnt=cnt+1;
ch = 'A';
if cnt==1 || cnt==2 || cnt==6 || cnt==5
    img = imabsdiff(imB,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='B';
    end
   img = imabsdiff(imC,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='C';
    end
    img = imabsdiff(imD,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='D';
    end
    img = imabsdiff(imE,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='E';
    end
    img = imabsdiff(imF,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='F';
    end
    img = imabsdiff(imG,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='G';
    end
    img = imabsdiff(imH,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='H';
    end
```

```
img = imabsdiff(imI,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='I';
end
img = imabsdiff(imJ,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='J';
end
img = imabsdiff(imK,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='K';
end
img = imabsdiff(imL,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='L';
end
img = imabsdiff(imM,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='M';
end
img = imabsdiff(imN,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='N';
end
img = imabsdiff(imO,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='O';
end
img = imabsdiff(imP,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='P';
end
img = imabsdiff(imQ,imgTest);
```

```
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='Q';
end
img = imabsdiff(imR,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='R';
end
img = imabsdiff(imS,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='S';
end
img = imabsdiff(imT,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='T';
end
img = imabsdiff(imU,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='U';
end
img = imabsdiff(imV,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='V';
end
img = imabsdiff(imW,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='W';
end
img = imabsdiff(imX,imgTest);
if(sum(img(:))<minDiff)
minDiff= sum(img(:));
ch='X';
end
img = imabsdiff(imY,imgTest);
if(sum(img(:))<minDiff)
```

```
    minDiff= sum(img(:));
    ch='Y';
    end
    img = imabsdiff(imZ,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='Z';
    end
else
    img = imabsdiff(im0,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='0';
    end
    img = imabsdiff(im1,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='1';
    end
    img = imabsdiff(im2,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='2';
    end
    img = imabsdiff(im3,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='3';
    end
    img = imabsdiff(im4,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='4';
    end
    img = imabsdiff(im5,imgTest);
    if(sum(img(:))<minDiff)
    minDiff= sum(img(:));
    ch='5';
    end
    img = imabsdiff(im6,imgTest);
    if(sum(img(:))<minDiff)
```

```
        minDiff= sum(img(:));
        ch='6';
        end
        img = imabsdiff(im7,imgTest);
        if(sum(img(:))<minDiff)
        minDiff= sum(img(:));
        ch='7';
        end
        img = imabsdiff(im8,imgTest);
        if(sum(img(:))<minDiff)
        minDiff= sum(img(:));
        ch='8';
        end
        img = imabsdiff(im9,imgTest);
        if(sum(img(:))<minDiff)
        minDiff= sum(img(:));
        ch='9';
        end
      end
      out=strcat(out,ch);
    end
end

msgbox(strcat('Vehicle Registraction Number :',out));
```

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